Reproduction of arXiv:1708.02243: Table I and Figures 2–7

Knapen, Lou, Papucci, Setford, Phys. Rev. D 96, 115015 (2017) · report of 21 October 2026 · Part A: the paper's literal method · Part B: deviations we tested · Table II: expected events at 300 fb^{-1}

Summary

- **How this report is organised. Part A** uses only the method written in the paper. **Part B** collects the changes we tried ourselves (our additional study), each with the reason. **Table II** (after Part B) lists the expected numbers of signal and background events.
- **Part A, Table I.** Trigger efficiency: colored 0.24 (paper 0.24, agrees), EW 0.18 (paper 0.10, does not agree). Fiducial efficiency: agrees for both. Reconstruction efficiency with the paper's SCT z resolution of 1.16 mm: far below the paper (colored 2 keV: 0.05 vs 0.35; EW 4 keV: 0.16 vs 0.72).
- **Part A, Figure 6.** Largest mass reached (exclusion / discovery): colored 1497 / 1424 GeV (paper 1793 / 1707), EW 524 / 473 GeV (paper 762 / 688).
- **Part A, backgrounds.** No converted-photon event (0 of 9139) and no pile-up event (0 of 5000) passes the signal region, as the paper says.
- **Part B, no SCT z smearing (our additional study).** The signal efficiencies and the Figure 6 reach come close to the paper (colored 1702 / 1591 GeV, EW 792 / 701 GeV), but then 146 of 9139 converted photons also pass (about 10–48 events at 300 fb^{-1}). No single SCT z setting gives both the paper's signal and its zero background.
- **Figures 2–5 and 7.** Fig. 2: in our simulation |L| returns to zero at every second meeting; in the paper it does not. Figs. 3 and 4: the shapes agree. Fig. 5: see above. Fig. 7: our hits are faster than the paper's, and its dE/dx panel could not be reproduced.

Questions to answer

- **Q1. Why is the converted-photon upper limit the same for both cross sections?** → Section A7.
- **Q2. Figure 2: why does our |L| go to zero at every second meeting while the paper's does not?** → Section A3.
- **Q3. Which smearing was applied, and how?** → Section A1 (paper's method) and Section B1 (no SCT z smearing).
- **Q4. Which uncertainties are quoted?** → Formulas (below) and every table (k/n = ε ± σ).

Formulas used (in plain words)

Trigger efficiency: ε_{trig} = N(events with $p_T(Q\bar{Q}) > 200$ GeV) / N(matched events)

Fiducial efficiency: ε_{fid} = N(fiducial events) / N(triggered events) (colored: triggered events in which both quirk-hadrons are charged)

Reconstruction efficiency: ε_{reco} = N(events in the signal region) / N(fiducial events)

Total efficiency (paper, Eq. 12): $\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$

Expected signal events: $N = L \times \sigma \times \varepsilon \times f_{ch}$ = $L \times \sigma \times$ N(signal region) / N(matched), L = 300 fb^{-1}

f_{ch} = fraction of the triggered colored events in which both quirk-hadrons are charged (EW: $f_{ch} = 1$). σ = cross section of the sample.

Expected converted-photon events: $N = L \times \sigma_\gamma \times P_{conv} \times$ N(pass) / 9139, P_{conv} = 1%

P_{conv} = 1% is taken from the paper's own numbers ("~1 pb … drops to ~10 fb if we require that the photon converts in the beampipe").

Plane fit (paper, Eqs. 9–10): $T_{ij} = \Sigma_a x_i^a x_j^a$ / (N − 1); Δs^2 = smallest eigenvalue of T; Δw^2 = second eigenvalue of T

Δs measures the thickness of the plane through the hits; Δw measures the width of the strip in which the hits lie.

Signal region (paper, Sec. IV): a plane is found; 7 layers with 2 hits and 1 layer with ≥ 1 hit; Δw < 1 cm; Δs < 0.01 cm

Angular momentum of the pair (Fig. 2): $L = (x_1^* - x_2^*) \times (p_1^* - p_2^*) / 2$ in the pair's rest frame; impact parameter b = |L| / p*

Figure 6: exclusion where N = 3, discovery where N = 5 (no background, as in the paper)

Statistical uncertainty: $\sigma_\varepsilon = \sqrt{(\varepsilon (1 - \varepsilon) / n)}$; if k = 0: 95% upper limit = $1 - 0.05^{1/n}$

Decision: **consistent** with the paper if $|\varepsilon - \varepsilon_{paper}| \le 2\,\sigma_0$, $\sigma_0 = \sqrt{(\varepsilon_{paper} (1 - \varepsilon_{paper}) / n)}$

PART A — The paper's literal method

Everything in Part A uses only what the paper states: LO 0 + 1 jet MLM-matched samples, SCT z resolution 1.16 mm on signal and background, ε_{trig} = matched count with $p_T(Q\bar{Q}) > 200$ GeV, no K-factor, no N_Q factor. The paper's numbers are shown next to ours.

A1. Settings

Setting	Value used in Part A	Source
Signal samples	MadGraph5_aMC@NLO 3.7.3 + Pythia 8, LO, pp → QQ̄ + 0 or 1 jet, MLM-matched; 2000 events per sample	paper: “0 + 1 jet … matched using Pythia8”
Cross section σ	LO matched σ of the sample: colored 1800 GeV 0.3287 fb, EW 800 GeV 0.2373 fb; no K-factor	the paper gives no σ
N_Q factor	none (σ × 1)	“N_Q = 2” appears only in the titles of the paper's Fig. 6 panels and is never defined
Trigger	$p_T(Q\bar{Q}) > 200$ GeV; ε_{trig} = matched count	paper, Sec. IV
Fiducial	both quirks cross all 8 layers (4 pixel + 4 SCT) within t < 25 ns	paper, Sec. IV
Layer crossings	first crossing of each layer	paper: each quirk “will most likely intersect each layer only once”
Signal region	plane found; 7 × 2 hits + 1 × ≥ 1 hit; Δw < 1 cm; Δs < 0.01 cm	paper, Sec. IV
Luminosity	300 fb^{-1}	paper, Fig. 6
Annihilation	not simulated	the paper assumes none

Detector part	Formula applied to the true crossing point	Value
Pixel layers (4)	hit moved to the centre of its pixel: $x_{meas} = p \cdot \mathrm{round}(x / p)$	p = 50 μm (rφ); 250 μm (IBL z), 400 μm (other z)
SCT layers (4), rφ	$(r\varphi)_{meas} = r\varphi + \sigma_{r\varphi} \cdot g$, g = Gaussian random number (mean 0, width 1)	$\sigma_{r\varphi}$ = 34 μm (2 × 17 μm)
SCT layers (4), z	$z_{meas} = z + \sigma_z \cdot g$	**σ_z = 1.16 mm** (2 × 580 μm)
Beam spot	all hits of an event shifted by the same z_0, $z_0 = 45$ mm · g	45 mm
Neighbouring pixels	hits in adjacent pixels of one layer merged into one hit	—

From the paper (Sec. III A) and its Ref. [27]. Pixel rounding gives an RMS error p/√12 (14 μm for 50 μm, 115 μm for 400 μm). One common z_0 per event is our reading of “a uniform, gaussian smearing … to the z-coordinates of all the hits”. The same smearing is applied to signal and background. $\sigma_z = 0$ is tested in Part B.

A2. Table I (SCT z = 1.16 mm)

Colored 1800 GeV (2000 events generated, 873 after matching)

Λ (keV)	ε_{trig} = N($p_T(Q\bar{Q}) > 200$ GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig., both charged)	ε_{reco} = N(signal region) / N(fiducial), SCT z 1.16 mm	paper ε_{reco}	decision
1	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	**0/22 = 0.00 (95% UL 0.13)**	0.083	consistent |Δ| 0.083 ≤ 2σ₀ 0.118
2	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	**1/22 = 0.05 ± 0.04**	0.35	**inconsistent** |Δ| 0.305 > 2σ₀ 0.203
3	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	**2/22 = 0.09 ± 0.06**	0.59	**inconsistent** |Δ| 0.499 > 2σ₀ 0.210
5	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	**5/22 = 0.23 ± 0.09**	0.74	**inconsistent** |Δ| 0.513 > 2σ₀ 0.187
10	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	**4/22 = 0.18 ± 0.08**	0.58	**inconsistent** |Δ| 0.398 > 2σ₀ 0.210
paper	0.24	0.28	see column		
dec.	consistent |Δ| 0.003 ≤ 2σ₀ 0.029	consistent |Δ| 0.044 ≤ 2σ₀ 0.109			

EW 800 GeV (2000 events generated, 1045 after matching)

Λ (keV)	ε_{trig} = N($p_T(Q\bar{Q}) > 200$ GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig.)	ε_{reco} = N(signal region) / N(fiducial), SCT z 1.16 mm	paper ε_{reco}	decision
1	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	**2/62 = 0.03 ± 0.02**	0.11	consistent |Δ| 0.078 ≤ 2σ₀ 0.079
2	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	**8/62 = 0.13 ± 0.04**	0.41	**inconsistent** |Δ| 0.281 > 2σ₀ 0.125
3	186/1045 = 0.18 ± 0.01	61/186 = 0.33 ± 0.03	**8/61 = 0.13 ± 0.04**	0.65	**inconsistent** |Δ| 0.519 > 2σ₀ 0.122

Λ (keV)	ε_{trig} = N($p_T(Q\bar{Q})$ > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig.)	ε_{reco} = N(signal region) / N(fiducial), SCT z 1.16 mm	paper ε_{reco}	decision
4	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	**10/62 = 0.16 ± 0.05**	0.72	**inconsistent** |Δ| 0.559 > 2σ₀ 0.114
5	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	**12/62 = 0.19 ± 0.05**	0.74	**inconsistent** |Δ| 0.546 > 2σ₀ 0.111
paper	0.10	0.28	see column		
dec.	**inconsistent** |Δ| 0.078 > 2σ₀ 0.019	consistent |Δ| 0.053 ≤ 2σ₀ 0.066			

ε_{fid} is recomputed at every Λ. EW at 3 keV: 61 instead of 62 because one event (no. 127) could not be integrated at 3 keV; like every such failure it is counted as not fiducial. For k = 0 the one-sided 95% upper limit is given. Decision cells give $|\Delta| = |\varepsilon - \varepsilon_{paper}|$ and $2\sigma_0$.

What we learn. With the paper's method: colored ε_{reco} is inconsistent at Λ = 2, 3, 5, 10 keV, consistent at the other Λ; EW ε_{reco} is inconsistent at Λ = 2, 3, 4, 5 keV, consistent at the other Λ. Where events pass, our values are 3–8 times lower than the paper's. ε_{trig} agrees for colored and is too high for EW (0.18 vs 0.10); ε_{fid} agrees for both.

Event counts at every Λ (cut-flow, first crossing)

Colored 1800 GeV: matched 873 → triggered 207 → both charged 68 (same for every Λ).

Λ (keV) → [SCT z 1.16 mm]	1	2	3	5	10
fiducial (both quirks cross all 8 layers, t < 25 ns)	22	22	22	22	22
plane found (Δw < 1 cm)	1	9	15	19	22
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	0	4	5	9	8
+ Δs < 0.01 cm = signal region (pass)	**0**	**1**	**2**	**5**	**4**
fail (fiducial − pass)	22	21	20	17	18
ε_{reco} = pass / fiducial	0.00	0.05	0.09	0.23	0.18
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$, $\times 10^{-2}$	**0.00**	**0.35**	**0.70**	**1.74**	**1.39**
paper ε $\times 10^{-2}$ (product of its Table I entries)	0.56	2.35	3.96	4.97	3.90

EW 800 GeV: matched 1045 → triggered 186 (same for every Λ).

Λ (keV) → [SCT z 1.16 mm]	1	2	3	4	5
fiducial (both quirks cross all 8 layers, t < 25 ns)	62	62	61	62	62
plane found (Δw < 1 cm)	14	36	40	45	49
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	7	21	21	24	24
+ Δs < 0.01 cm = signal region (pass)	**2**	**8**	**8**	**10**	**12**
fail (fiducial − pass)	60	54	53	52	50
ε_{reco} = pass / fiducial	0.03	0.13	0.13	0.16	0.19
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$, $\times 10^{-2}$	**0.19**	**0.77**	**0.77**	**0.96**	**1.15**
paper ε $\times 10^{-2}$ (product of its Table I entries)	0.31	1.15	1.82	2.02	2.07

A3. Figure 2: angular momentum L of the quirk pair (with Appendix A of arXiv:2002.07503)

Equations used (the paper's model, solved numerically):

Eq. 3: $d(m\gamma\mathbf{v})/dt = -\Lambda^2 [\sqrt{(1-v_\perp^2)}\,\hat{s} + (v_\parallel/\sqrt{(1-v_\perp^2)})\,\mathbf{v}_\perp] + q\,\mathbf{v} \times \mathbf{B}$ (string + magnetic force)

$\mathbf{L} = (\mathbf{x}_1^* - \mathbf{x}_2^*) \times (\mathbf{p}_1^* - \mathbf{p}_2^*) / 2$ in the pair's rest (CM) frame at equal CM time; $d = |\mathbf{x}_1 - \mathbf{x}_2|$; r = distance of the pair's centre from the beam

Eq. 5: in the pair's rest frame the magnetic field acts as an electric field $\mathbf{E}^* = \gamma_{cm}\,\mathbf{v}_{cm} \times \mathbf{B}$; torque $\boldsymbol{\tau} = q\,\mathbf{d} \times \mathbf{E}^*$ (paper: $\tau \sim 2\mathbf{d} \times e\mathbf{E}_{cm}$, d = one quirk's distance from the CM)

What Appendix A of arXiv:2002.07503 shows (Li, Li, Pei, Zhang, "The quirk trajectory", same equation, uniform magnetic field):

- Eqs. (II.11)–(II.14): in the pair's rest frame the two quirks move back and forth on one line and meet at every integer g ($g = \Lambda^2 t/(2m\rho)$ is the time in units of half an oscillation).
- Text after Eq. (A.4): "the average torque on the system of two quirks is zero when g is increased from 2n to 2n + 2 such that the total angular momentum of the quirk pair is also invariant at g = 2n".
- Eq. (A.5): $z_1(2n) = -z_2(2n) = 0$, i.e. at every second meeting the quirks meet head-on, as at production.

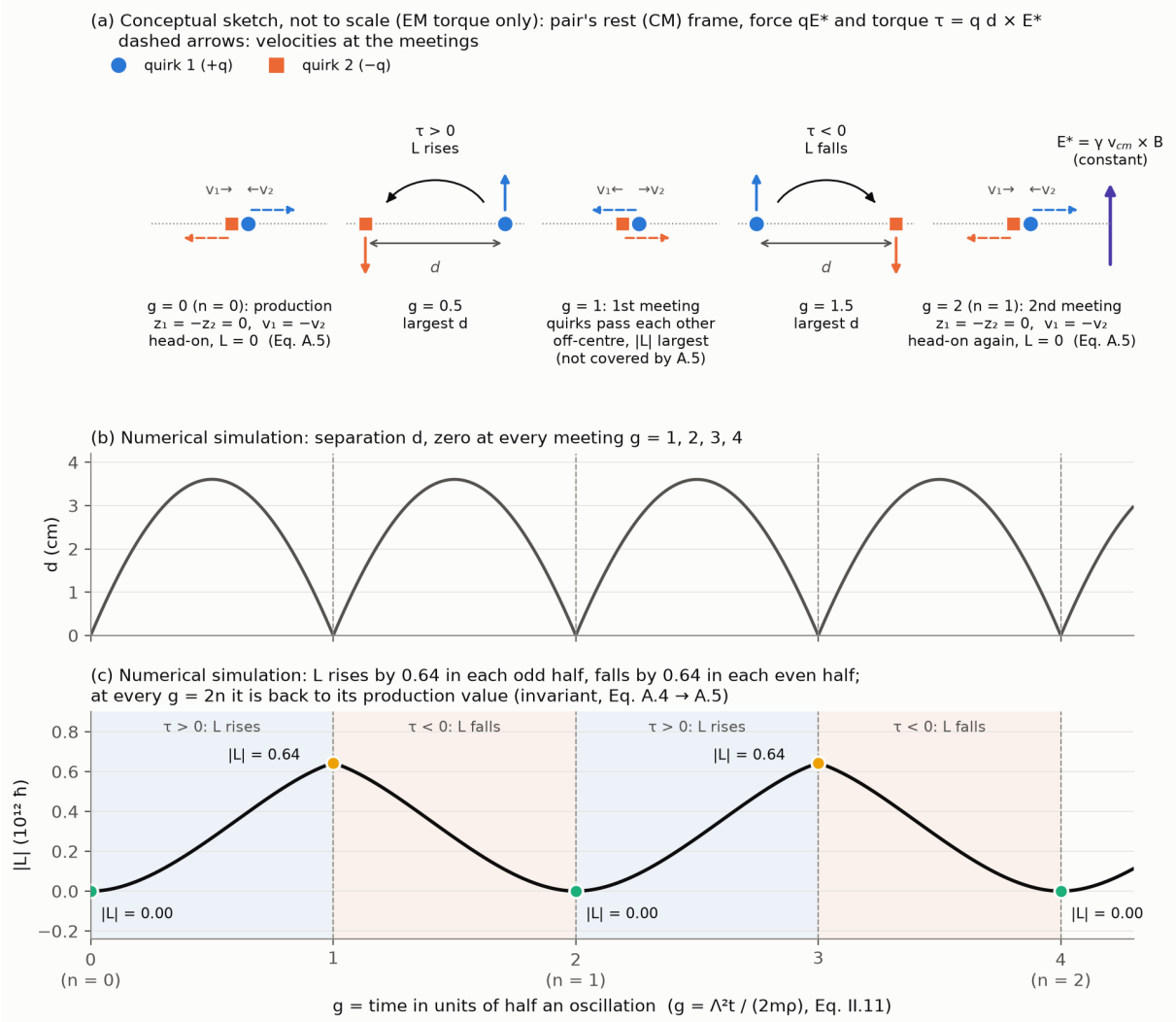


Figure: how L builds up and returns. (a) is a **conceptual sketch** (not to scale, EM torque only; not simulation output) of the mechanism and of the relation of Eq. A.5 at the head-on meetings g = 0 and 2: $z_1 = -z_2 = 0$ with opposite velocities $v_1 = -v_2$. Eq. A.5 does not cover the odd meetings, where the quirks pass off-centre with the largest |L|. (b) and (c) are the **numerical simulation** of Eq. 3 with the Lorentz force for the event with the paper's kinematics: $|L| = 0.64 \times 10^{12}$ ħ at g = 1, 3 and consistent with zero at g = 2, 4. The agreement with Eq. A.4–A.5 is the numerical result; (a) only illustrates it.

Why. Between two meetings the separation **d** points one way, in the next half-oscillation the other way. The field **E*** is constant, so the torque q**d**×**E*** has the same size and the opposite sign in the two halves: L goes up, then back to zero. Our simulation shows exactly this.

Change of L in each half-oscillation (event with the paper's kinematics, units 10^8 ħ)	**1st**	**2nd**	**3rd**	**4th**
total change of L in our simulation	+6434	-6430	+6426	-6429
from the magnetic force (torque q**d**×**E***)	+5563	-5559	+5557	-5559
from the string's own term in Eq. 3	+871	-870	+870	-869
paper's Fig. 2 (read from its curves)	**+6452**	**-8390**	**+7075**	**-8007**

In our simulation each half-oscillation adds or removes the same amount ($\pm 0.64 \times 10^{12}$ ħ = $\pm 6400 \times 10^8$ ħ), so L returns to zero after every two halves. In the paper the halves are unequal (+0.65, −0.84, +0.71, −0.80, in 10^{12} ħ), so L never returns to zero at a meeting.

m = 1800 GeV, Λ = 2 keV, B = 2 T along the beam, production at the origin, RK4 with 3000 steps per oscillation.

Initial values of our two simulated events	**Event with the paper's kinematics**	**One event of our sample (no. 8)**
charges	+2/3, −2/3	+1, -1
quirk 1 lab momentum (p_x, p_y, p_z)	212.2 , 0.0 , 1204.1 GeV	138.4 , -1123.0 , -424.7 GeV
quirk 2 lab momentum	212.2 , 0.0 , -1204.1 GeV	51.5 , 1430.4 , 144.6 GeV
pair speed β_{cm} (transverse)	0.0975	0.0808
internal speed v* (CM frame)	0.556	0.5874
p* = m γ* v*	1204.1 GeV	1307.0 GeV
predicted d max = 2(γ* − 1) m/Λ²	3.607 cm	4.185 cm

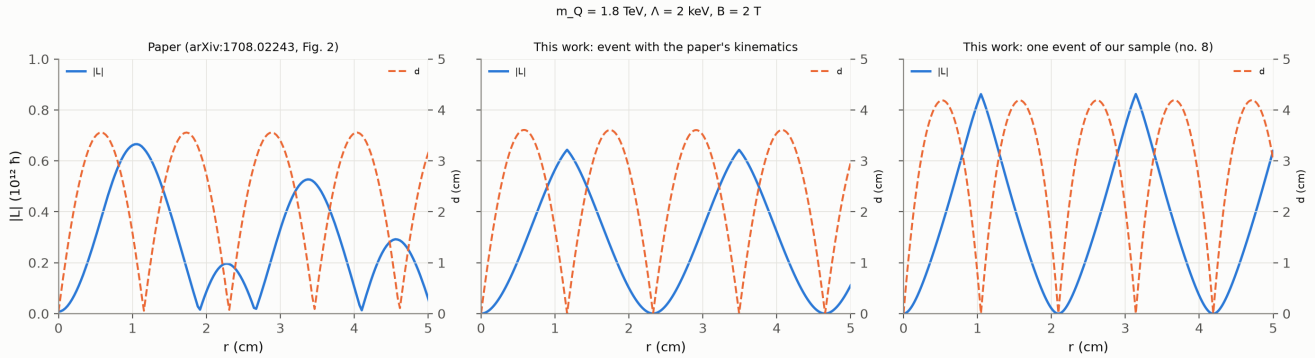
Initial values of our two simulated events	Event with the paper's kinematics	One event of our sample (no. 8)
where the values come from	v* from the paper's d max (Eq. 4), β_{cm} from the spacing of its meetings; oscillation along the beam and charges ±2/3 chosen by us (the paper does not give them)	MadGraph 3.7.3 + Pythia 8.317 colored sample; chosen because its d max and spacing are nearest the paper's; charges of the final-state R-hadrons

Meeting by meeting (paper: from its vector figure; its signed L assumes a sign change at each zero of |L|)

meeting		r (cm)	|L| (10^8 ħ)	signed L (10^8 ħ)	change of L in this half (10^8 ħ)	|L| = 0 at r (cm)	b = |L| / p* (cm)
1	**paper**	1.16	6477	+6477	**+6400**	—	1.1×10^{-5}
	ours, paper's kinematics	1.16	6435	+6435	**+6430**	—	1.1×10^{-5}
	ours, event no. 8	1.05	8621	+8621	**+8620**	—	1.3×10^{-5}
2	**paper**	2.31	1946	−1946	**-8420**	between meetings: 1.91	3.2×10^{-6}
	ours, paper's kinematics	2.33	4.2	±4.2	**-6430**	at the meeting	6.9×10^{-9}†
	ours, event no. 8	2.10	1.4	±1.4	**-8620**	at the meeting	2.1×10^{-9}†
3	**paper**	3.46	5150	+5150	**+7100**	between meetings: 2.67	8.4×10^{-6}
	ours, paper's kinematics	3.49	6431	+6431	**+6430**	—	1.1×10^{-5}
	ours, event no. 8	3.14	8623	+8623	**+8620**	—	1.3×10^{-5}
4	**paper**	4.62	2875	−2875	**-8020**	between meetings: 4.10	4.7×10^{-6}
	ours, paper's kinematics	4.65	1.3	±1.3	**-6430**	at the meeting	2.2×10^{-9}†
	ours, event no. 8	4.19	4.9	±4.9	**-8620**	at the meeting	7.4×10^{-9}†

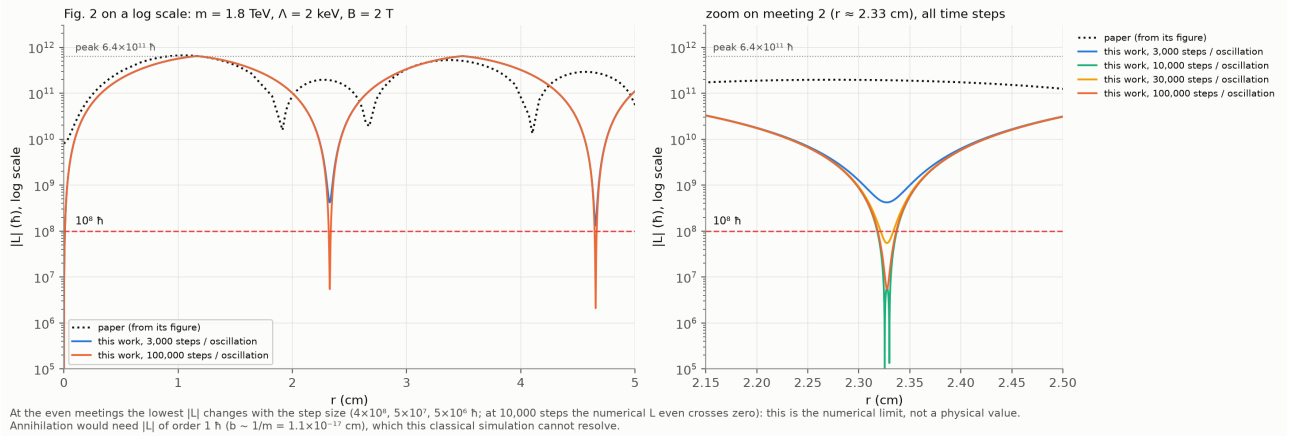
d max is the same in every half-oscillation for the paper (3.556 cm) and for us (3.61 cm, no energy loss). b uses p* = 1204 GeV for the paper (it gives no p*); $1/m_Q = 1.1\times10^{-17}$ cm. † At the even meetings our left-over |L| (1–5 in units of 10^8 ħ) is the limit of the simulation and changes with the time step; it is not converged. The paper's first-half change is +6400 in this table (from its curve's first point, 80) and +6452 in the half-oscillation table above (from zero); the difference is the reading precision of its curve.

What the table shows. The first meeting agrees (0.65 vs 0.64 $\times10^{12}$ ħ). In our simulation every half-oscillation changes L by the same amount (±0.64$\times10^{12}$ ħ; ±0.86$\times10^{12}$ ħ for event 8), so |L| is back to zero at every second meeting. In the paper the changes are unequal (+0.64, −0.84, +0.71, −0.80): L changes sign between the meetings, its large peaks fall (0.67 → 0.53) and its small ones rise (0.20 → 0.29; all in 10^{12} ħ). With the same d in every half and a uniform field, the paper's Eq. 3 and Lorentz force give equal changes; the paper does not state what else entered its calculation.



Conclusion. Our Fig. 2 is what the paper's own equation gives; the same result is found analytically in arXiv:2002.07503 (Appendix A) and with an independent code. The paper's later |L| values cannot be obtained from its stated equation; the paper does not give the event's charges and momenta or how L was computed.

A3.1 Is L exactly zero? Fig. 2 on a log scale



At the even meetings the lowest |L| changes with the step size (4×10^8, 5×10^7, 5×10^6 ħ; at 10,000 steps the numerical L even crosses zero): this is the numerical limit, not a physical value. Annihilation would need |L| of order 1 ħ (b ~ 1/m = 1.1×10^{-11} cm), which this classical simulation cannot resolve.

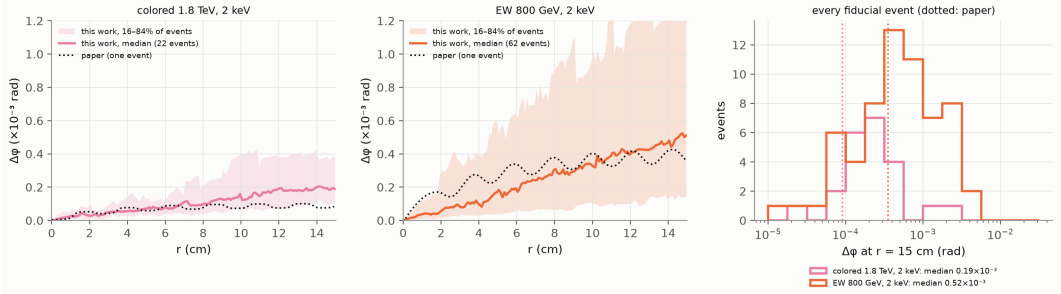
RK4 steps per oscillation	|L| at meeting 2 (10^8 ħ)	|L| at meeting 4 (10^8 ħ)	|L| at meetings 1 and 3 (10^8 ħ)
3,000	4.21	1.32	6430
10,000	0.059	0.237	6430
30,000	0.55	0.577	6430
100,000	0.0545	0.021	6430

What the log scale shows. At the meetings 1 and 3, |L| = 6.43×10^{11} ħ for every step size (converged, physical). At the meetings 2 and 4 the left-over |L| changes with the step size ($4\times10^8 \to 5\times10^7 \to 5\times10^6$ ħ; at 10,000 steps it even crosses zero): it is the numerical limit, not a physical value. So: on the scale of Fig. 2 (10^{12} ħ) L returns to zero, as Appendix A of arXiv:2002.07503 predicts exactly; at the ħ level needed for annihilation the classical simulation cannot decide. The paper's 2×10^{11} ħ at meeting 2 is about 460× the largest of our values.

A4. Figure 3: turning of the quirks' plane

Δφ = angle between the plane of the two velocity vectors at radius r and at production (paper's definition)

Fig. 3: Δφ at r = 15 cm (Λ = 2 keV)	fiducial events	ours: median [16–84%]	paper (one event)	our events below the paper's: k / n = fraction ± σ	match
colored 1800 GeV	22	0.19 [0.10, 0.38] ×10⁻³ rad	0.09 ×10⁻³ rad	2 / 22 = 0.09 ± 0.06	**at the low end of our spread**
EW 800 GeV	62	0.52 [0.14, 1.70] ×10⁻³ rad	0.36 ×10⁻³ rad	24 / 62 = 0.39 ± 0.06	inside our 16–84% band

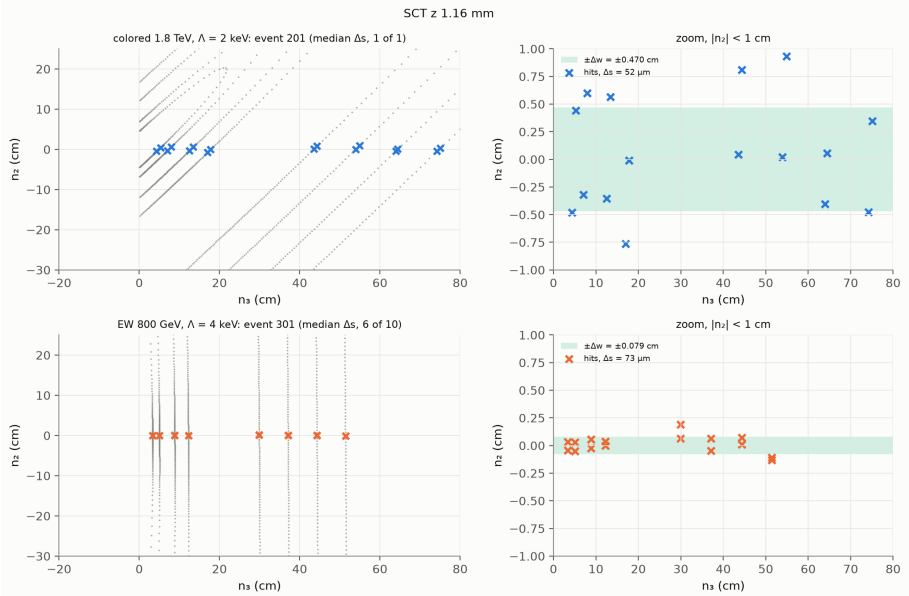


What we learn. The plane turns by less than about 10^{-3} rad over 15 cm in both samples, so the hits stay on one flat plane, as in the paper. The paper's single event lies inside our spread (colored: at its low end).

A5. Figure 4: hits of one signal event

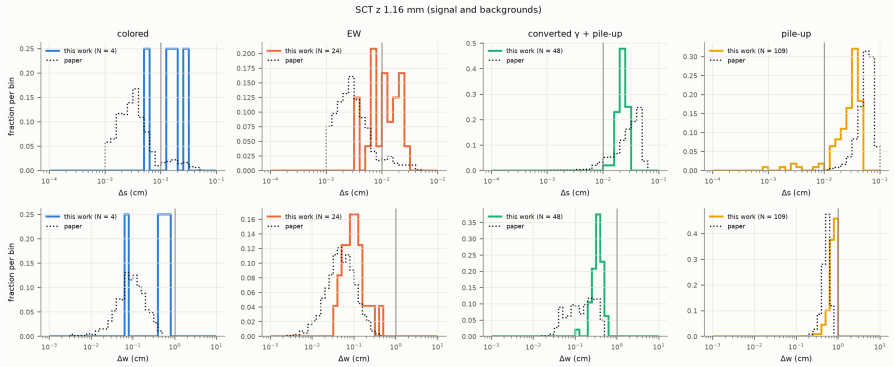
Fig. 4 example event	Δs (plane thickness)	Δw (strip width)	match with the paper
Paper (colored)	83 μm	0.074 cm	—
This work, colored, SCT z 1.16 mm	52 μm	0.470 cm	Δs yes (both below the 100 μm cut); Δw no (wider)
This work, EW, SCT z 1.16 mm	73 μm	0.079 cm	Δs yes (both below the 100 μm cut); Δw close

Each example is the signal event with the median Δs among the events in the signal region. All hits lie on a thin strip, as in the paper.



A6. Figure 5: Δs and Δw distributions (SCT z 1.16 mm on signal and background)

Fig. 5	Δs median: ours / paper	Δw median: ours / paper	fraction with Δs < 0.01 cm: ours / paper	decision
colored signal	160 / 32 μm	0.550 / 0.092 cm	1/4 = 0.25 ± 0.22 / 0.90	**inconsistent** |Δ| 0.649 > 2σ₀ 0.302
EW signal	110 / 28 μm	0.092 / 0.048 cm	10/24 = 0.42 ± 0.10 / 0.92	**inconsistent** |Δ| 0.500 > 2σ₀ 0.113
converted γ + pile-up	222 / 306 μm	0.350 / 0.156 cm	0/48 = 0.00 (95% UL 0.06) / 0.07	consistent |Δ| 0.069 ≤ 2σ₀ 0.073
pile-up	317 / 571 μm	0.778 / 0.525 cm	8/109 = 0.07 ± 0.02 / 0.00	**inconsistent** |Δ| 7.2×10^{-2} > 2σ₀ 7.6×10^{-3}



What we learn. With 1.16 mm no converted photon passes the Δs cut, as in the paper (paper 0.07), but most of the signal fails it too: only 0.25 (colored) and 0.42 (EW) of the signal planes have Δs < 0.01 cm, against about 0.9 in the paper. Our median Δw is 6× (colored) and 2× (EW) the paper's.

A7. Backgrounds (SCT z 1.16 mm)

Background sample	events	plane found (Δw < 1 cm)	+ hit multiplicity	pass (signal region)	fail	paper
converted γ + pile-up (μ = 50)	9139	1990	48	**0**	9139	“a few” events at 300 fb⁻¹, “can easily be removed”
pile-up only (μ = 50)	5000	31 (≥ 4 of 8 layers); 1 (all 8)	—	**0**	5000	none in the signal region

Pile-up planes (5000 events)	ours: k / n	rate	paper (approx.)	decision
selected plane, ≥ 1 hit in ≥ 4 of 8 layers	31 / 5000	6.2×10^{-3}	$\sim1\times10^{-3}$	**inconsistent** $\lvert\Delta\rvert$ $5.2\times10^{-3} > 2\sigma_0$ 8.9×10^{-4}
selected plane, ≥ 1 hit in all 8 layers	1 / 5000	2.0×10^{-4}	$\sim6\times10^{-5}$	consistent $\lvert\Delta\rvert$ $1.4\times10^{-4} \le 2\sigma_0$ 2.2×10^{-4}

Converted photons from Z(→νν) + γ (MadGraph, MET > 200 GeV), converted in the beam pipe, with pile-up μ = 50 overlaid.

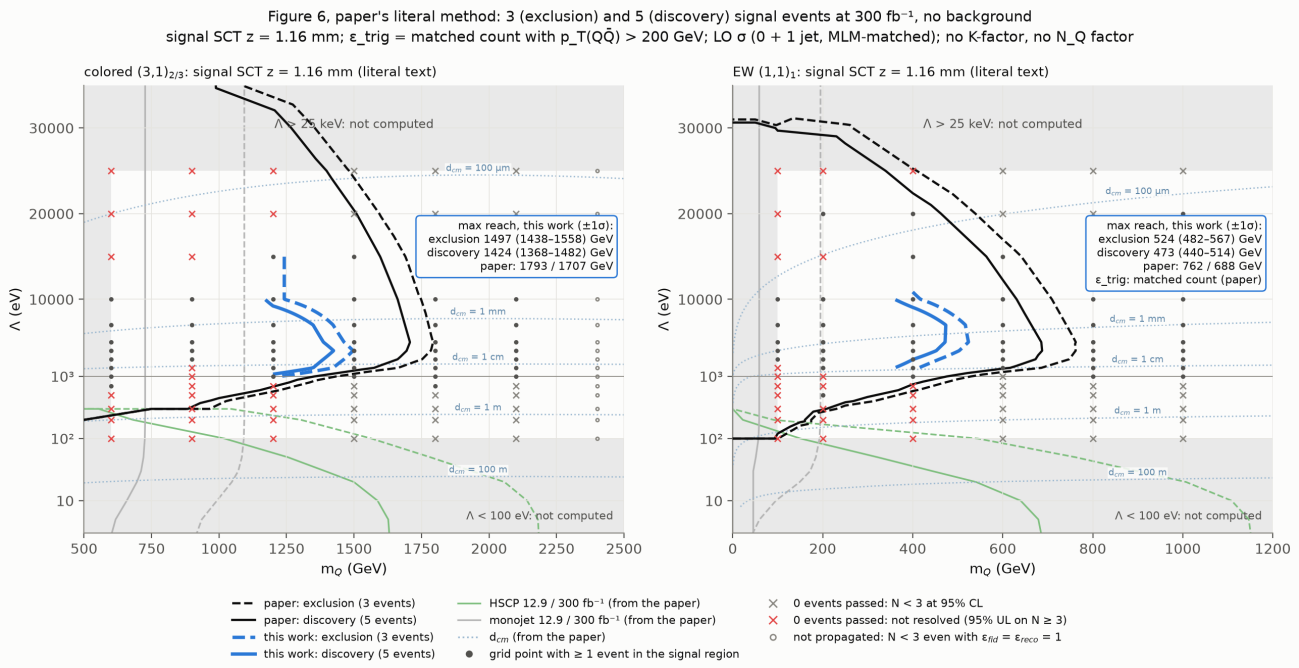
Converted photons: why the upper limit is the same for both cross sections (Q1)

Converted γ + pile-up (9139 events)	in signal region	per conversion	expected events at 300 fb⁻¹: our σ (0.211 pb) / paper σ (~1 pb)
SCT z 1.16 mm	0 / 9139	$< 3.3\times10^{-4}$ (95% UL)	< 0.21 / < 0.98

Answer. The limit is a property of our 9139 simulated conversions: 0 pass, so the 95% upper limit per conversion is $1 - 0.05^{1/9139} = 3.3\times10^{-4}$, whatever the cross section. The cross section only sets how many conversions are expected (300 fb⁻¹ × σ × 1% = 633 or 3000), which is why the last column differs.

A8. Figure 6: discovery and exclusion contours (paper's method)

- **Paper:** contours of 3 (exclusion) and 5 (discovery) signal events at 300 fb^{-1}, "assuming negligible irreducible backgrounds"; $N = L\ \sigma\ \varepsilon_{trig}\ \varepsilon_{fid}\ \varepsilon_{reco}$.
- **This work:** 13 samples (MG5 3.7.3 + Pythia8, MLM-matched, 2000 events per mass: colored 600–2400 GeV, EW 100–1000 GeV) through the Table I pipeline at Λ = 0.1–25 keV, signal SCT z 1.16 mm. σ = LO matched σ of each sample; ε_{trig} = matched count; colored includes the both-charged factor f_{ch}; no K-factor; no N_Q factor.
- EW ε_{trig} at 800 GeV in the Fig. 6 sample: 152/1024 = 0.148 ± 0.011. Fig. 6 uses this value, not Table I's 186/1045 = 0.178 ± 0.012, because each Fig. 6 point takes all three efficiencies from its own sample. The two samples are independent generations with identical generator settings and the same trigger definition ($p_T(QQ) > 200$ GeV); only the random seed differs. The two values differ by 1.8σ; both are far above the paper's 0.10.
- **Not computed:** $\Lambda > 25$ keV (the integrator's 20-million-step ceiling removes 5–56% of the fiducial events at 30–35 keV), $\Lambda < 100$ eV, colored < 600 GeV, EW < 100 GeV; low-Λ edges at light masses are not resolved (0 events passed, red ×).



Largest mass reached (GeV), ±1σ	colored exclusion	colored discovery	EW exclusion	EW discovery
paper, Fig. 6	1793	1707	762	688
paper, abstract	–	1600	–	650
this work, paper's method (SCT z 1.16 mm)	1497 (1438–1558)	1424 (1368–1482)	524 (482–567)	473 (440–514)
difference from the paper's figure	-17%	-17%	-31%	-31%

What we learn. With the paper's method our contours have the paper's shape (lower edge near 1 keV, largest mass at $\Lambda \approx$ 4–7 keV), but they reach less far: colored -17% / -17%, EW -31% / -31% (exclusion / discovery). The reason is the low ε_{reco} with SCT z 1.16 mm (Section A2).

A9. Figure 7: dE/dx and βγ of the pixel hits (paper's Appendix A)

- **Paper:** “For the dE/dx we use the most probable value as a function of βγ [PDG] … we omitted hits with βγ < 0.1 in the left-hand panel”, benchmarks EW 800 GeV and colored 1800 GeV at Λ = 2 keV. Silicon thickness and the event/hit selection are not stated.
- **This work:** βγ = |p|/m at each pixel-layer crossing (production integrator, Fig. 6 samples at these masses); dE/dx = PDG Eq. 34.12 (Landau–Vavilov–Bichsel most probable value) with the silicon density effect, IBL 200 µm and 250 µm for the other pixel layers (our choice); |z| = 1. Check: 1.11 MeV cm²/g for a minimum-ionising particle in 250 µm (PDG ≈ 1.1).

Figure 7 (Appendix A): pixel hits, Λ = 2 keV; our bins are not the paper's (paper bin width not stated), so compare shapes and ranges

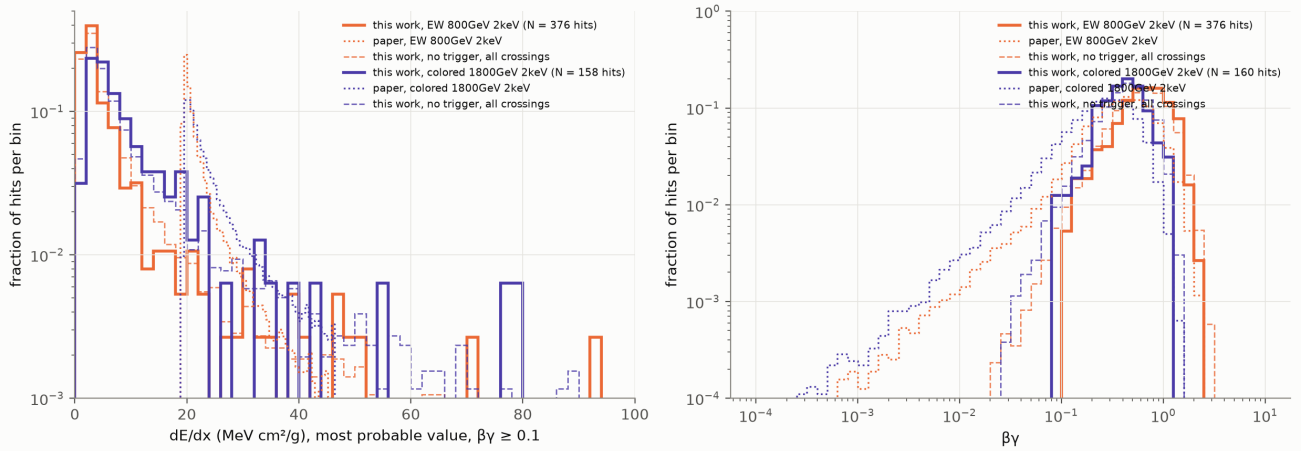


Fig. 7 (pixel hits, Λ = 2 keV)	hits	βγ median	hits with βγ < 0.1	dE/dx median (MeV cm²/g)	dE/dx range of the histogram
paper, EW 800 GeV	–	0.35	0.07–0.10	5.6 (PDG MPV of its own βγ histogram)	19–47
this work, EW 800 GeV: triggered, fiducial, first crossing, t < 25 ns	376	0.66	0.000	3.1	1–93
all matched events (no trigger), all crossings, t < 25 ns	8558	0.62	0.012		
all matched events, all crossings, no timing cut	9428	0.61	0.012		
paper, colored 1800 GeV	–	0.22	0.16–0.22	12.6 (PDG MPV of its own βγ histogram)	19–47
this work, colored 1800 GeV: triggered, fiducial, first crossing, t < 25 ns	160	0.42	0.013	6.2	2–79
all matched events (no trigger), all crossings, t < 25 ns	2626	0.43	0.022		
all matched events, all crossings, no timing cut	3059	0.44	0.023		

What we learn. Both distributions peak at βγ ≈ 0.1–1, but our hits are faster: median βγ 0.66 / 0.42 (EW / colored) against the paper's 0.35 / 0.22. Three differences: (1) the paper has a tail of very slow hits (βγ down to 10^{-4}; 7%–10% (EW) and 16%–22% (colored) of its hits below βγ = 0.1, the range given by the bin that contains 0.1); we find at most a few % in any selection (with or without trigger, timing cut or first-crossing requirement). (2) Because our hits are faster, our dE/dx is lower. (3) The paper's dE/dx histogram lies at 19–47 MeV cm²/g for both samples, while the PDG most probable value of its own βγ histogram spans 1–47 with a median of about 6 (EW) and 13 (colored) MeV cm²/g; ours (median 3 and 6) lies in that range and is lower because our hits are faster. The paper does not give enough detail (formula, thickness, units, selection) to trace either point.

A9.1 Test: which dE/dx definition does the paper use?

The paper's own βγ histogram (right panel) is converted to dE/dx with four definitions (silicon, density effect, |z| = 1; 50 points per βγ bin; bins below βγ = 0.1, where the Bethe formula does not hold, left out: 7% (EW) and 16% (colored) of the paper's hits) and compared with the paper's dE/dx histogram (left panel). **Criterion, fixed before the test:** for both samples, median within 15% of the paper's, and the 10% and 90% quantiles each within 25%.

Definition	EW 800: 10% / median / 90%	colored 1800: 10% / median / 90%	hits below 18.9 (EW / colored)	passes
PDG MPV, MeV cm²/g	2.1 / 6.3 / 32.4	3.5 / 12.1 / 50.3	80% / 65%	no
Bethe mean, MeV cm²/g	2.7 / 7.0 / 27.4	4.3 / 12.2 / 38.9	82% / 68%	no
Bethe mean × ρ, MeV/cm	6.3 / 16.4 / 63.8	9.9 / 28.3 / 90.6	56% / 33%	no
PDG MPV × ρ, MeV/cm	4.8 / 14.6 / 75.3	8.2 / 28.1 / 117.1	58% / 36%	no

Definition	EW 800: 10% / median / 90%	colored 1800: 10% / median / 90%	hits below 18.9 (EW / colored)	passes
paper's dE/dx histogram	19.5 / 20.7 / 28.2	19.9 / 22.4 / 33.4	0 (starts at 18.9)	

Result. No definition passes. The paper's dE/dx histogram is narrow and starts sharply at 18.9 (10% quantile 19.5–19.9), while every conversion of its own βγ histogram puts 33%–82% of the hits below 18.9 and is several times broader. The mean Bethe loss in MeV/cm comes closest in median (ratio 0.79 / 1.26) but not in shape or range. So none of the four standard definitions explains the left panel from the right one, the same holds with the βγ cut at 0.05, and the statement above stands: the paper does not give enough detail to trace its dE/dx axis.

A10. Quirks crossing the same layer more than once

Sample (fiducial events, all crossings within 25 ns)	events with > 2 hits on at least one layer	fraction ± σ
Colored 1800 GeV, 2 keV	13 / 22	0.59 ± 0.10
EW 800 GeV, 4 keV	31 / 62	0.50 ± 0.06

These are results of our simulation. The paper says each quirk “will most likely intersect each layer only once”. arXiv:2002.07503 (Sec. VI, Table I and Appendix D, Table II) also finds repeated crossings, more often at larger Λ and larger m/p_T, with 4–8 hits on some layers at Λ = 2 keV; its percentages (~10%) are for other masses and the CMS tracker and are not directly comparable. Part A uses the first crossing only, as the paper implies.

What Part A shows. With only the paper's stated settings we reproduce the trigger efficiency for colored quirks, the fiducial efficiencies, the flat planes (Figs. 3 and 4), and a background-free signal region. We do not reproduce the reconstruction efficiency (3–8 times lower), the EW trigger efficiency (0.18 vs 0.10), Figure 2 after the first meeting and Figure 7. Because ε_{reco} is low, the Figure 6 reach is also lower than the paper's.

PART B — Deviations we tested (our additional study)

Each deviation changes Part A in one respect. They are not part of the paper's method; they show which change would bring our numbers closer to the paper's.

Deviation	Why we tested it (one line)	Section
No SCT z smearing ($\sigma_z = 0$, signal and background)	With the paper's 1.16 mm our ε_{reco} is far below Table I; we checked whether a better z resolution reproduces the paper.	B1
LO-ratio trigger	Our EW ε_{trig} (0.18) is far above the paper's 0.10; arXiv:1911.02223 computes the trigger as an LO cross-section ratio, so we tried that.	B2
σ scaling (K-factor)	The paper gives no cross sections; scaling σ shows how much the Fig. 6 reach depends on the normalisation.	B3
$N_Q = 2$	The paper's Fig. 6 titles say "$N_Q = 2$ Quirks" without a definition; we tried reading it as a factor 2 on σ.	B4
Colored without f_{ch}	The paper quotes ~30% for both quirk-hadrons charged but does not say whether this factor enters Fig. 6.	B5
Annihilation	Our |L| returns to zero at every second meeting, so the quirks might annihilate; we estimated a worst case.	B6

B1. No SCT z smearing on signal and background

Everything as in Part A, but $\sigma_z = 0$ in the SCT for the signal and for the backgrounds. This is our additional study.

Colored 1800 GeV

Λ (keV)	ε_{reco}, no SCT z smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm (Part A)
1	**2/22 = 0.09 ± 0.06**	0.083	consistent |Δ| 0.008 ≤ 2σ₀ 0.118	0/22 = 0.00 (95% UL 0.13)
2	**8/22 = 0.36 ± 0.10**	0.35	consistent |Δ| 0.014 ≤ 2σ₀ 0.203	1/22 = 0.05 ± 0.04
3	**17/22 = 0.77 ± 0.09**	0.59	consistent |Δ| 0.183 ≤ 2σ₀ 0.210	2/22 = 0.09 ± 0.06
5	**20/22 = 0.91 ± 0.06**	0.74	consistent |Δ| 0.169 ≤ 2σ₀ 0.187	5/22 = 0.23 ± 0.09
10	**14/22 = 0.64 ± 0.10**	0.58	consistent |Δ| 0.056 ≤ 2σ₀ 0.210	4/22 = 0.18 ± 0.08

EW 800 GeV

Λ (keV)	ε_{reco}, no SCT z smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm (Part A)
1	**12/62 = 0.19 ± 0.05**	0.11	**inconsistent** |Δ| 0.084 > 2σ₀ 0.079	2/62 = 0.03 ± 0.02
2	**40/62 = 0.65 ± 0.06**	0.41	**inconsistent** |Δ| 0.235 > 2σ₀ 0.125	8/62 = 0.13 ± 0.04
3	**41/61 = 0.67 ± 0.06**	0.65	consistent |Δ| 0.022 ≤ 2σ₀ 0.122	8/61 = 0.13 ± 0.04
4	**50/62 = 0.81 ± 0.05**	0.72	consistent |Δ| 0.086 ≤ 2σ₀ 0.114	10/62 = 0.16 ± 0.05
5	**42/62 = 0.68 ± 0.06**	0.74	consistent |Δ| 0.063 ≤ 2σ₀ 0.111	12/62 = 0.19 ± 0.05

What we learn. Without SCT z smearing: colored ε_{reco} is consistent with the paper at every Λ; EW ε_{reco} is inconsistent at Λ = 1, 2 keV, consistent at the other Λ (at 1 and 2 keV ours is higher than the paper's). So the paper's Table I is reproduced only if the SCT z position is measured much better than 1.16 mm.

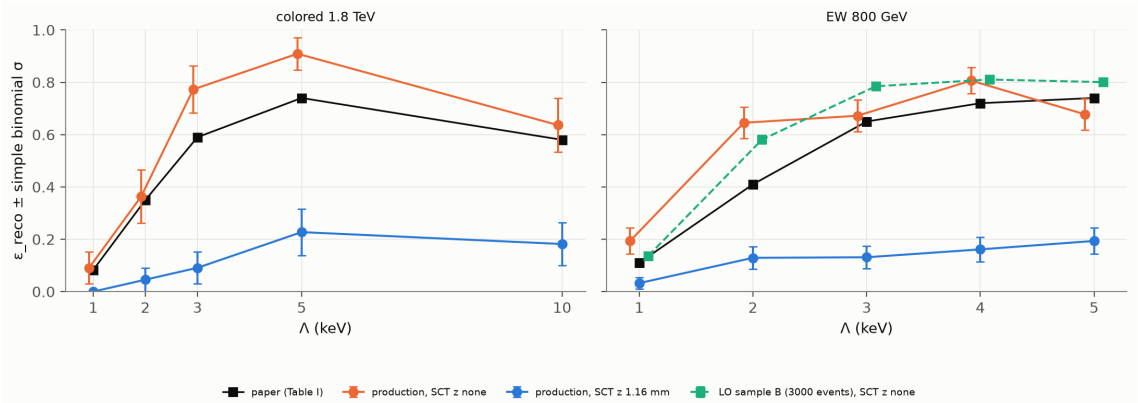
Cut-flow without SCT z smearing

Colored 1800 GeV: matched 873 → triggered 207 → both charged 68 (same for every Λ).

Λ (keV) → [no SCT z smearing]	1	2	3	5	10
fiducial (both quirks cross all 8 layers, t < 25 ns)	22	22	22	22	22
plane found (Δw < 1 cm)	4	18	21	22	22
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	2	8	17	20	14
+ Δs < 0.01 cm = signal region (pass)	**2**	**8**	**17**	**20**	**14**
fail (fiducial − pass)	20	14	5	2	8
ε_{reco} = pass / fiducial	0.09	0.36	0.77	0.91	0.64
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$, $\times 10^{-2}$	**0.70**	**2.79**	**5.93**	**6.97**	**4.88**
paper ε $\times 10^{-2}$ (product of its Table I entries)	0.56	2.35	3.96	4.97	3.90

EW 800 GeV: matched 1045 → triggered 186 (same for every Λ).

Λ (keV) → [no SCT z smearing]	**1**	**2**	**3**	**4**	**5**
fiducial (both quirks cross all 8 layers, t < 25 ns)	62	62	61	62	62
plane found (Δw < 1 cm)	21	54	61	62	62
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	12	40	41	50	42
+ Δs < 0.01 cm = signal region (pass)	**12**	**40**	**41**	**50**	**42**
fail (fiducial − pass)	50	22	20	12	20
ε_{reco} = pass / fiducial	0.19	0.65	0.67	0.81	0.68
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$, $\times 10^{-2}$	**1.15**	**3.83**	**3.92**	**4.78**	**4.02**
paper ε $\times 10^{-2}$ (product of its Table I entries)	0.31	1.15	1.82	2.02	2.07



ε_{reco} vs Λ: the paper, our production samples with SCT z none and 1.16 mm, and the EW leading-order sample (Section B2), all with simple binomial σ.

Figures 4 and 5 without SCT z smearing

Fig. 4 example event	**Δs (plane thickness)**	**Δw (strip width)**	**match with the paper**
Paper (colored)	83 μm	0.074 cm	—
This work, colored, no SCT z smearing	38 μm	0.352 cm	Δs yes (both below the 100 μm cut); Δw no (wider)
This work, EW, no SCT z smearing	45 μm	0.100 cm	Δs yes (both below the 100 μm cut); Δw close

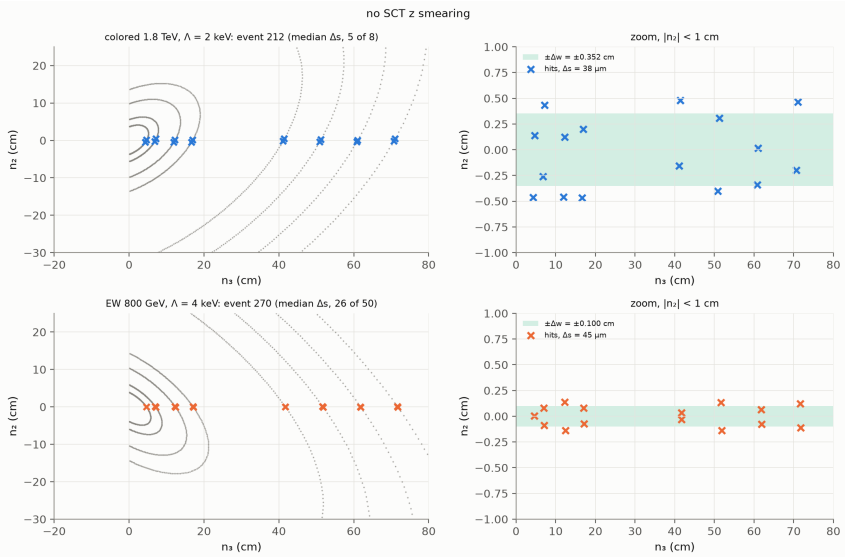
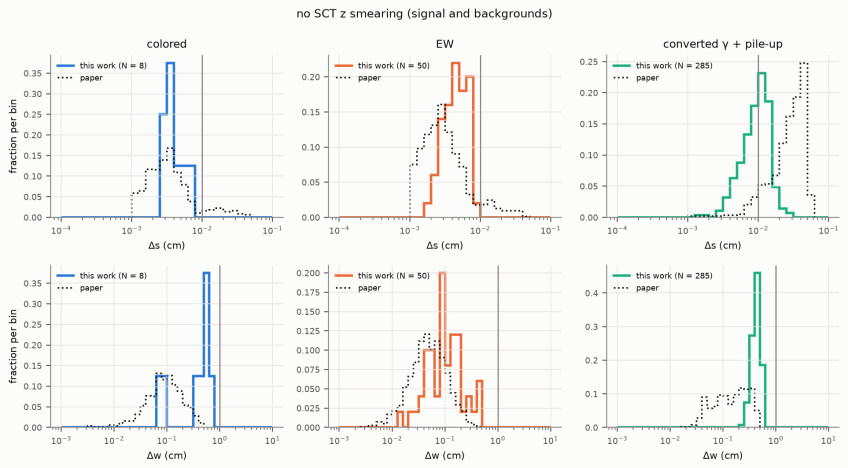


Fig. 5	**Δs median: ours / paper**	**Δw median: ours / paper**	**fraction with Δs < 0.01 cm: ours / paper**	**decision**
colored signal	38 / 32 μm	0.482 / 0.092 cm	8/8 = 1.00 (95% LL 0.69) / 0.90	consistent $\lvert\Delta\rvert$ 0.101 ≤ 2σ₀ 0.213
EW signal	44 / 28 μm	0.097 / 0.048 cm	50/50 = 1.00 (95% LL 0.94) / 0.92	**inconsistent** $\lvert\Delta\rvert$ 0.084 > 2σ₀ 0.078
converted γ + pile-up	97 / 306 μm	0.438 / 0.156 cm	146/285 = 0.51 ± 0.03 / 0.07	**inconsistent** $\lvert\Delta\rvert$ 0.443 > 2σ₀ 0.030

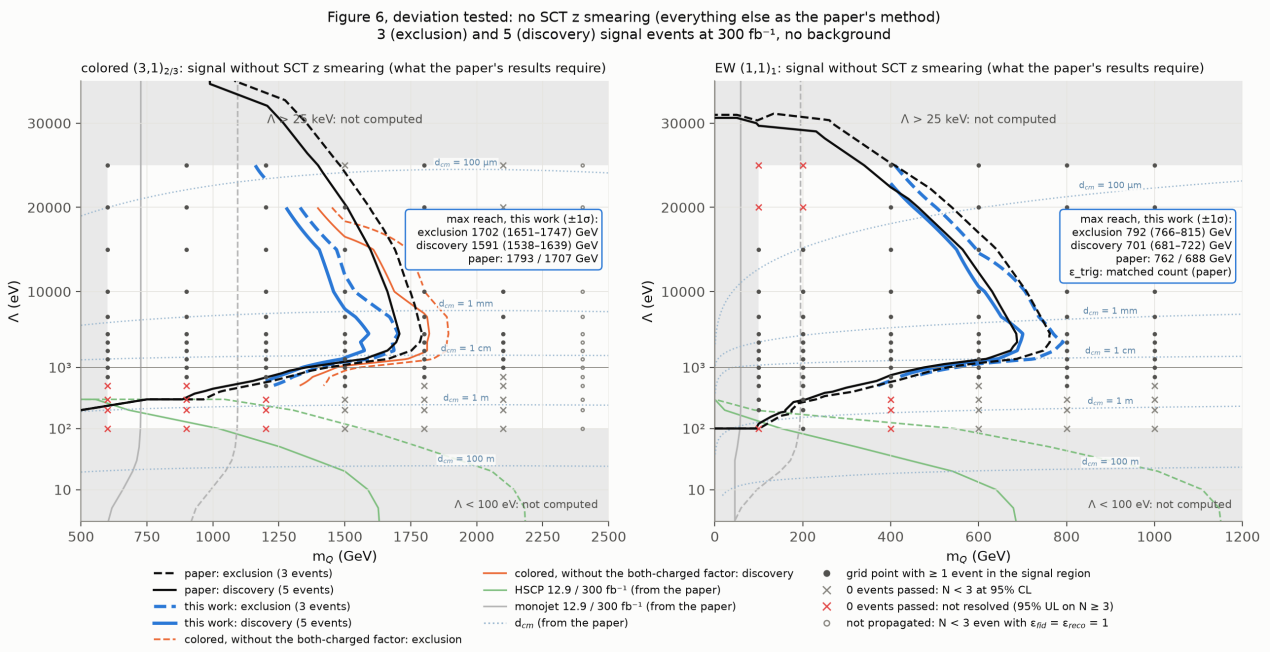


Pile-up only was not simulated without SCT z smearing.

Background (9139 converted γ + pile-up)	pass	fraction	expected events at 300 fb⁻¹: our σ (0.211 pb) / paper σ (~1 pb)
no SCT z smearing	146 / 9139	0.0160 ± 0.0013	10 / 48
SCT z 1.16 mm (Part A)	0 / 9139	< 3.3×10^{-4} (95% UL)	< 0.21 / < 0.98

What we learn. Without smearing the signal Δs (≈ 40 μm) is close to the paper's (≈ 30 μm) and all signal passes the Δs cut (paper ≈ 90%), but 0.51 of the converted-photon planes also pass (paper 0.07), i.e. 146 of 9139, about 10 (our σ) to 48 (paper's σ) events at 300 fb⁻¹. With 1.16 mm no converted photon passes, but most signal fails too. **No single SCT z setting reproduces both the paper's signal and its background.**

Figure 6 without SCT z smearing



Largest mass reached (GeV), ±1σ	colored exclusion	colored discovery	EW exclusion	EW discovery
paper, Fig. 6	1793	1707	762	688
this work, no SCT z smearing	1702 (1651–1747)	1591 (1538–1639)	792 (766–815)	701 (681–722)
difference from the paper's figure	-5%	-7%	+4%	+2%
this work, SCT z 1.16 mm (Part A)	1497 (1438–1558)	1424 (1368–1482)	524 (482–567)	473 (440–514)

Without SCT z smearing the converted-photon background is not zero (above), so these contours, drawn without background as in the paper, are optimistic.

B2. LO-ratio trigger (EW)

Why tested: our EW ε_{trig} (0.18, matched count) is far above the paper's 0.10; arXiv:1911.02223 obtains about 0.1 as a leading-order cross-section ratio. Two leading-order samples were made for the EW benchmark; everything else as in Part A but without SCT z smearing.

Reconstruction efficiency ε_{reco} = N(signal region) / N(fiducial), no SCT z smearing, first crossing

Λ (keV)	Production: MadGraph + Pythia, matched (2000 generated)	LO, no shower (3000 events)	LO + Pythia shower (2792 events)	paper ε_{reco}
1	12/62 = 0.19 ± 0.05 **inconsistent** |Δ| 0.084 > 2σ₀ 0.079	121/885 = 0.14 ± 0.01 **inconsistent** |Δ| 0.027 > 2σ₀ 0.021	127/950 = 0.13 ± 0.01 **inconsistent** |Δ| 0.024 > 2σ₀ 0.020	0.11
2	40/62 = 0.65 ± 0.06 **inconsistent** |Δ| 0.235 > 2σ₀ 0.125	520/895 = 0.58 ± 0.02 **inconsistent** |Δ| 0.171 > 2σ₀ 0.033	548/955 = 0.57 ± 0.02 **inconsistent** |Δ| 0.164 > 2σ₀ 0.032	0.41
3	41/61 = 0.67 ± 0.06 consistent |Δ| 0.022 ≤ 2σ₀ 0.122	707/901 = 0.78 ± 0.01 **inconsistent** |Δ| 0.135 > 2σ₀ 0.032	750/953 = 0.79 ± 0.01 **inconsistent** |Δ| 0.137 > 2σ₀ 0.031	0.65
4	50/62 = 0.81 ± 0.05 consistent |Δ| 0.086 ≤ 2σ₀ 0.114	727/897 = 0.81 ± 0.01 **inconsistent** |Δ| 0.090 > 2σ₀ 0.030	779/959 = 0.81 ± 0.01 **inconsistent** |Δ| 0.092 > 2σ₀ 0.029	0.72
5	42/62 = 0.68 ± 0.06 consistent |Δ| 0.063 ≤ 2σ₀ 0.111	719/898 = 0.80 ± 0.01 **inconsistent** |Δ| 0.061 > 2σ₀ 0.029	746/961 = 0.78 ± 0.01 **inconsistent** |Δ| 0.036 > 2σ₀ 0.028	0.74

Trigger and fiducial efficiency

	Production	LO, no shower	LO + Pythia shower	paper
ε_{trig}	186/1045 = 0.18 ± 0.01 **inconsistent** |Δ| 0.078 > 2σ₀ 0.019	σ_B/σ_A = 0.096 ± 0.001 consistent |Δ| 0.004 ≤ max(2σ, 0.005)	0.096 × 2792/3000 = 0.089 ± 0.001 **inconsistent** |Δ| 0.011 > max(2σ, 0.005)	0.10
ε_{fid} (4 keV)	62/186 = 0.33 ± 0.03 consistent |Δ| 0.053 ≤ 2σ₀ 0.066	897/3000 = 0.30 ± 0.01 **inconsistent** |Δ| 0.019 > 2σ₀ 0.016	959/2792 = 0.34 ± 0.01 **inconsistent** |Δ| 0.063 > 2σ₀ 0.017	0.28

For the cross-section ratios the statistical error (±0.001) is smaller than the precision of the paper's value (0.10, two decimals), so these two cells use |Δ| ≤ max(2σ, 0.005); all other cells use the 2σ₀ rule of the Formulas section.

How the two leading-order samples were made

- **Why.** Our trigger efficiency for the EW benchmark (0.18) is far above the paper's 0.10. arXiv:1911.02223 (Appendix A) obtains a trigger efficiency of about 0.1 as a leading-order cross-section ratio, σ(pp → QQ̄ + jet, p_T(jet) > 200 GeV) / σ(pp → QQ̄). The paper does not state how its value was obtained, so we computed the EW benchmark in this way.
- **Generation.** MadGraph5_aMC@NLO 3.7.3, quirk model, pp at 13 TeV, PDF nn23lo1, m_Q = 800 GeV. Sample A: pp → QQ̄, σ_A = 0.2307 fb. Sample B: pp → QQ̄ + jet with p_T(jet) > 200 GeV, σ_B = 0.02214 fb, 3000 unweighted events. $\varepsilon_{trig} = \sigma_B/\sigma_A$ = 0.096 ± 0.001; MadGraph 2.3.3 (the version in the paper's source) gives 0.096.
- **LO, no shower:** the quirk momenta are taken directly from the 3000 events of sample B. **LO + shower:** the same events are showered with Pythia 8.317 (no matching); 2792 of 3000 keep p_T(QQ̄) > 200 GeV after the shower, so $\varepsilon_{trig} = \sigma_B/\sigma_A$ × 2792/3000.
- **Everything else is as for the production sample:** detector model, Λ values, trajectory simulation, fiducial definition, plane search and signal region, first crossing, no SCT z smearing. ε_{fid} = N(fiducial) / N(events in the sample).
- **Numerical failures:** 122 of 15000 event–Λ trajectories (LO) and 111 of 13960 (LO + shower) could not be integrated; they are counted as not fiducial, as in the production sample.

Result. Without shower the leading-order ratio gives ε_{trig} = 0.096, consistent with the paper's 0.10; after the shower 2792/3000 events keep p_T(QQ̄) > 200 GeV and ε_{trig} = 0.089 (inconsistent). So the paper's 0.10 corresponds to the leading-order ratio, not to our matched sample (0.18). EW ε_{trig} at 800 GeV in the Fig. 6 sample: 152/1024 = 0.148 ± 0.011. Fig. 6 uses this value, not Table I's 186/1045 = 0.178 ± 0.012, because each Fig. 6 point takes all three efficiencies from its own sample. The two samples are independent generations with identical generator settings and the same trigger definition (p_T(QQ̄) > 200 GeV); only the random seed differs. The two values differ by 1.8σ; both are far above the paper's 0.10. ε_{reco} is almost the same in all three samples (0.58 / 0.57 at 2 keV, 0.81 / 0.81 at 4 keV, LO / LO + shower) and is above the paper at Λ = 1, 2, 3, 4, 5 keV in the LO sample. With about 800–900 fiducial events the uncertainty is ±0.01–0.02, so this is not a statistical fluctuation, and it does not depend on how the events were generated. For the colored benchmark the same ratio gives 0.14; the matched production sample already agrees with the paper (0.24).

Figure 6 with the LO-ratio trigger (GeV)	SCT z 1.16 mm: EW excl. / disc.	no SCT z smearing: EW excl. / disc.
EW ε_{trig} = matched count (Part A)	524 / 473	792 / 701
EW ε_{trig} = LO ratio (0.096 at 800 GeV)	474 / 428	712 / 643
LO-ratio trigger and ε_{fid}, ε_{reco} from the LO sample at 800 GeV	–	717 / 648
paper, Fig. 6	762 / 688	762 / 688

The LO ratio lowers EW ε_{trig} by a factor 0.54–0.75 (100–1000 GeV). The paper does not state this method.

B3. σ scaling (K-factor)

All cross sections multiplied by K (largest mass reached in GeV; not a fit). K = 1 is Part A (left) and B1 (right).

Variant	SCT z 1.16 mm: colored excl. / disc.	SCT z 1.16 mm: EW excl. / disc.	no smearing: colored excl. / disc.	no smearing: EW excl. / disc.
σ × 1 (our LO)	1497 / 1424	524 / 473	1702 / 1591	792 / 701
σ × 0.8	–	501 / 454	–	750 / 670
σ × 0.9	–	513 / 464	–	772 / 686
σ × 1.2	–	542 / 490	–	824 / 730
σ × 1.4	–	558 / 506	–	850 / 759
σ × 1.5	1553 / 1482	–	1789 / 1679	–
σ × 1.8	1578 / 1507	–	1817 / 1718	–
σ × 2	1593 / 1522	594 / 542	1831 / 1741	910 / 824
paper, Fig. 6	1793 / 1707	762 / 688	1793 / 1707	762 / 688

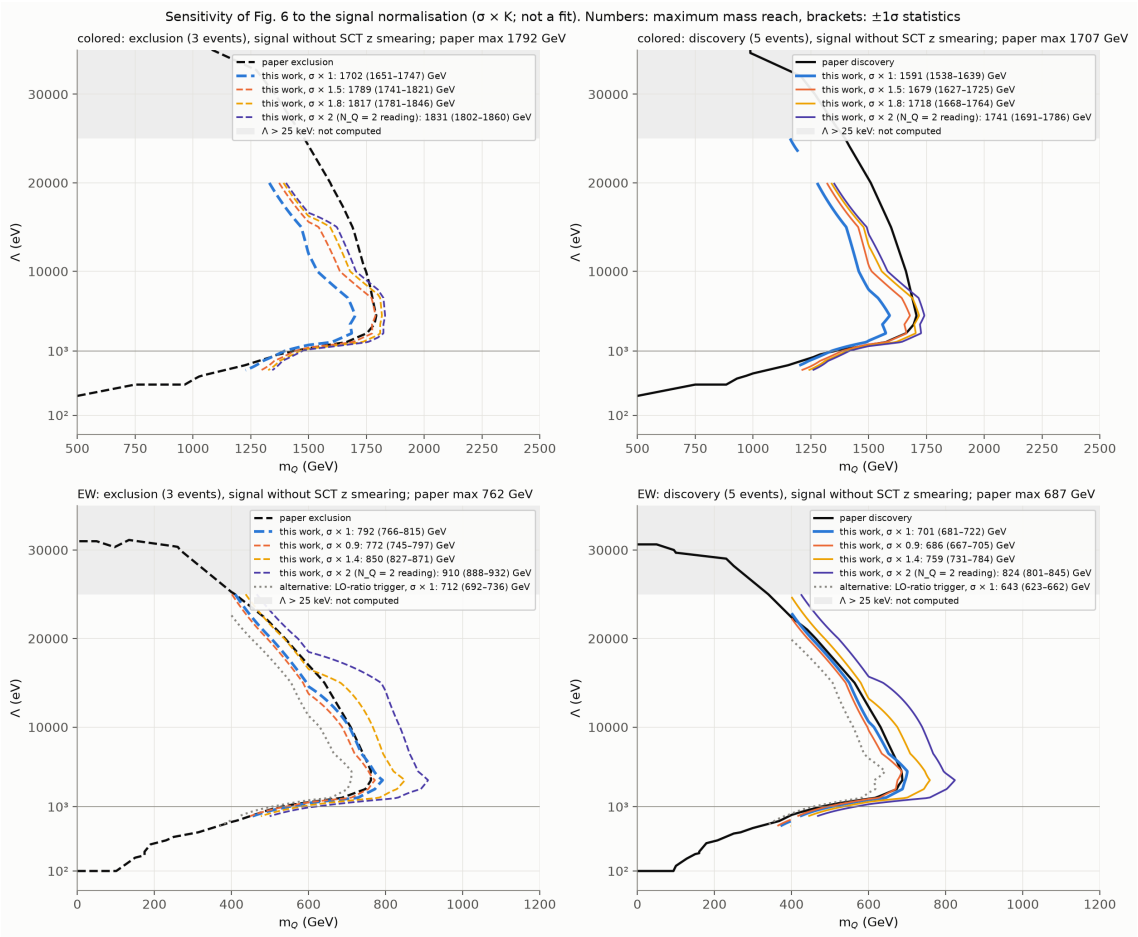


Figure: no SCT z smearing, σ × K. Dotted grey (EW): LO-ratio trigger (B2).

What we learn. Without SCT z smearing, EW matches the paper's figure with K ≈ 0.85–0.9, while colored needs K ≈ 1.5–1.7, so one common NLO K-factor cannot explain both. With SCT z 1.16 mm even σ × 2 stays below the paper. The paper gives no σ, so the origin cannot be identified.

B4. $N_Q = 2$ read as a factor 2 on σ

"$N_Q = 2$" appears only in the titles of the paper's Fig. 6 panels (arXiv v1 and v2) and is never defined or used in the text. It could mean two quirk species (a factor 2 on σ) or only the two quirks of each pair (no factor). The row σ × 2 of the table above is this reading.

B5. Colored without the both-charged factor f_{ch}

Colored, largest mass reached (GeV), ±1σ	SCT z 1.16 mm: excl.	SCT z 1.16 mm: disc.	no smearing: excl.	no smearing: disc.
with f_{ch} (Part A, B1)	1497 (1438–1558)	1424 (1368–1482)	1702 (1651–1747)	1591 (1538–1639)
without f_{ch}	1655 (1592–1724)	1586 (1527–1651)	1893 (1859–1929)	1820 (1784–1852)
paper, Fig. 6	1793	1707	1793	1707

f_{ch} = 68/207 = 0.33 at 1800 GeV (paper, Sec. II: ~30%). The paper's ε_{fid} for colored quirks is defined for triggered events with both quirk-hadrons charged, so f_{ch} is needed to turn its ε into a number of events; the paper does not say whether its Fig. 6 includes it.

B6. Annihilation: does L = 0 mean the quirks annihilate?

Why tested: our |L| returns to zero at every second meeting (Section A3), so the quirks might annihilate; the paper assumes they do not.
Paper (Sec. II, text of Fig. 2): “whenever |L| = 0, the separation between the quirks is large”, so they do not annihilate. **This work and arXiv:2002.07503 (App. A):** |L| returns to its production value at every second meeting. So the paper's reason does not hold. Two points show why this still does not mean annihilation:

B6.1 Classical L ≈ 0 is not a collision

- Classically L = b × p, where b is the distance by which the two quirks miss each other (impact parameter): b = |L| / p.
- To annihilate, the quirks must overlap at the quantum level, b ~ $1/m_Q$ = 1.1×10^{-17} cm for m_Q = 1800 GeV. At that distance L is only about one ħ (b > $1/m_Q$ needs |L| > 0.67 ħ).
- In our simulation (and in the independent code) the left-over |L| at the even meetings is 10^6–10^8 ħ, so the quirks still miss each other by millions of times $1/m_Q$:

time steps per oscillation	|L| at meeting 2 (10^8 ħ)	|L| at meeting 4 (10^8 ħ)	b = |L| / p (cm)	b / ($1/m_Q$)
3,000	4.21	1.32	2×10^{-9} – 7×10^{-9}	2×10^{8} – 6×10^{8}
10,000	0.059	0.237	1×10^{-10} – 4×10^{-10}	9×10^{6} – 4×10^{7}
30,000	0.55	0.577	9×10^{-10} – 9×10^{-10}	8×10^{7} – 9×10^{7}
100,000	0.0545	0.021	3×10^{-11} – 9×10^{-11}	3×10^{6} – 8×10^{6}

Event with the paper's kinematics, p* = 1204 GeV. A classical simulation cannot resolve L of a few ħ: the left-over L changes with the time step and does not converge, so we do not know where it would stop.

B6.2 Even if L returns exactly to its production value, annihilation is not certain

- arXiv:2002.07503 (App. A, Eq. A.5): at every second meeting the quirks meet head-on, as at production, where the pair's angular momentum is zero (arXiv:2108.06748, Sec. 4: “the initial L'_{tot} at the IP is zero”). With this symmetry annihilation becomes **possible**.
- Possible is not the same as happening. The annihilation cross section of heavy quirks is small, $\sigma v \propto \alpha^2/m_Q^2$ (arXiv:2404.13814, Eq. 4.11). Worst case, if an annihilation chance ≈ $\alpha^2\gamma v$ is allowed at every second meeting (α = 0.02 / 0.05), the fraction of pairs surviving inside the tracker is 0.995 / 0.970 (colored, 2 keV), 0.971 / 0.835 (EW, 4 keV) and 0.887 / 0.494 (colored, 10 keV): small at the benchmarks, not small at large Λ.
- The symmetry is not perfect in a real detector. arXiv:2108.06748 (Sec. 4) shows that ionisation in material gives the pair angular momentum, and L changes sharply when the pair enters or leaves material, or when only one quirk is inside it. Tracker layers are material. Radiation and a non-uniform field are also absent from the paper's model and ours. (The turning angle η of arXiv:2002.07503, Eq. A.6, rotates the oscillation but keeps the head-on meetings of Eq. A.5, so η alone does not stop L from returning.)
- arXiv:2404.13814 models annihilation only after the pair has radiated its energy and dropped into the quirkonium ground state, not during the oscillations. It finds that radiative energy loss matters only for Λ ≳ 100 keV, far above the keV values here.

Conclusion. L returning to zero (this work, Appendix A) removes the paper's argument, but it does not show that the quirks annihilate. That needs a quantum estimate of the per-meeting probability, which neither the paper nor this study provides. Part A, like the paper, assumes no annihilation.

Variant	SCT z 1.16 mm: colored excl. / disc.	SCT z 1.16 mm: EW excl. / disc.	no smearing: colored excl. / disc.	no smearing: EW excl. / disc.
annihilation, worst case (α = 0.05, chance at every second meeting)	1479 / 1407	495 / 450	1672 / 1560	758 / 671
paper, Fig. 6	1793 / 1707	762 / 688	1793 / 1707	762 / 688

Effect on Figure 6 of the worst case above (largest mass reached in GeV).

Table II. Expected events at 300 fb^{-1}

N = L × σ × N(pass) / N(sample), L = 300 fb^{-1}. Two detector settings, each applied to signal **and** background: SCT z 1.16 mm (the paper's method, Part A) and no SCT z smearing (Part B1). For k = 0 the 95% upper limit is given. The uncertainty is statistical (binomial on N(pass) / N(sample)); the MadGraph σ error (≈ 2–3%) is not included.

Process	Λ (keV)	σ (fb)	SCT z 1.16 mm: pass / sample	SCT z 1.16 mm: N	no SCT z smearing: pass / sample	no SCT z smearing: N	N with the paper's ε (our σ)
colored 1800 GeV	1	0.3287	0 / 873	**< 0.34**	2 / 873	**0.23 ± 0.16**	0.18
	2		1 / 873	**0.11 ± 0.11**	8 / 873	**0.90 ± 0.32**	0.76
	3		2 / 873	**0.23 ± 0.16**	17 / 873	**1.92 ± 0.46**	1.28
	5		5 / 873	**0.56 ± 0.25**	20 / 873	**2.26 ± 0.50**	1.61
	10		4 / 873	**0.45 ± 0.23**	14 / 873	**1.58 ± 0.42**	1.26
EW 800 GeV	1	0.2373	2 / 1045	**0.14 ± 0.10**	12 / 1045	**0.82 ± 0.23**	0.22
	2		8 / 1045	**0.54 ± 0.19**	40 / 1045	**2.72 ± 0.42**	0.82
	3		8 / 1045	**0.54 ± 0.19**	41 / 1045	**2.79 ± 0.43**	1.30
	4		10 / 1045	**0.68 ± 0.21**	50 / 1045	**3.41 ± 0.47**	1.44
	5		12 / 1045	**0.82 ± 0.23**	42 / 1045	**2.86 ± 0.43**	1.48
converted γ + pile-up, our σ	–	211 × 1% conv.	0 / 9139	**< 0.21**	146 / 9139	**10.12 ± 0.83**	“a few”, removable
converted γ + pile-up, paper's σ	–	1000 × 1% conv.	0 / 9139	**< 0.98**	146 / 9139	**47.93 ± 3.93**	“a few”, removable
pile-up only (μ = 50)	–	not given	0 / 5000	< 6.0×10^{-4} per event	not simulated	–	none

Signal: N(sample) = matched events; N(pass) = events in the signal region (for colored these are among the triggered events with both quirk-hadrons charged, so f_{ch} is included). Last column: the paper's ε = ε_{trig} ε_{fid} ε_{reco} (Table I) × our σ × our f_{ch} (colored 0.33; EW 1). Converted photons: σ after MET > 200 GeV (ours 0.211 pb, paper ~1 pb) × 1% conversion probability. Pile-up only: the paper gives no number of triggered events that the pile-up sits on, so only the probability per event is given.

What Table II shows. With SCT z 1.16 mm on both, the background is zero (upper limit < 0.2–1 event), but the signal is also small: at most 0.8 events at these benchmarks. Without SCT z smearing on both, the signal is 3–9 times larger, but the converted-photon background (10 or 48 events) is then larger than the signal. At these benchmark masses (1800 and 800 GeV) the signal is below or near the 3-event exclusion level in both settings, consistent with Figure 6, where these masses lie at or beyond the reach (Figure 6 uses its own samples, e.g. EW 800 GeV ε_{trig} = 0.148 instead of 0.178).

Why the paper uses Δs < 0.01 cm and Δw < 1 cm

- **What they measure.** Δs is the thickness of the plane through the hits (smallest eigenvalue of T); Δw is the width of the strip in which the hits lie (second eigenvalue). The two quirks move in one plane, so their hits form a thin, narrow strip; random hits do not.
- **Δw < 1 cm: reason given.** Sec. IV: “We deliberately do not impose a tight cut on Δw, as the efficiency for such a cut is strongly signal dependent. The rather loose cut of Δw < 1.0 cm is intended to retain decent efficiencies for quirks with larger oscillation amplitude (low Λ).” The same 1 cm is used in the seeding and in the event selection (Sec. III C), and the paper attributes the low-Λ drop of ε_{reco} “largely” to “a stringent Δw requirement”.
- **Δs < 0.01 cm: no derivation given.** It appears only as “A signal selection cut Δs < 0.01 cm is indicated on the left figure” (Fig. 5), i.e. it is read off Fig. 5, between the signal (≈ 30 μm) and the backgrounds (hundreds of μm). The scale is the expected one: the tracker resolution is “roughly ~ 100 μm” (Sec. I). Looser windows are used earlier: seeding Δs < 0.05 cm and $|x \cdot n_1|$ < 0.05 cm, growth factor 3 (Sec. III C).
- **General statement** (Sec. III C): “A tighter selection will generally boost computational efficiency at a cost of reduced signal efficiency, which is what has been chosen in this work.”
- **Our numbers:** signal Δs median 110–160 μm with SCT z 1.16 mm (Part A) and 38–44 μm without smearing (Part B); backgrounds 100–600 μm. The value of the cut is not the issue; the SCT z resolution decides whether the signal passes it.

Open points (what is still unclear)

- **SCT z resolution.** With the stated 1.16 mm our ε_{reco} is far below Table I; only no SCT z smearing reproduces it, and then the converted-photon background is not zero.
- **EW trigger efficiency.** Our matched count gives 0.18 (Fig. 6 sample: 0.15); the paper's 0.10 is reproduced only by an LO cross-section ratio, which the paper does not mention.
- **Cross sections.** The paper gives no signal σ and no K-factor; “N_Q = 2” is not defined.

- **Converted-photon normalisation.** Our σ after MET > 200 GeV is 0.211 pb, the paper's ~1 pb (factor 4.7, not explained); the 1% conversion probability is derived from the paper's own numbers.
- **Pile-up.** No absolute normalisation is given in the paper; our rate of planes with hits in ≥ 4 of 8 layers is about 6× the paper's.
- **Δw.** Our signal Δw is 2–5 times the paper's.
- **Figure 2.** The paper's later |L| values cannot be obtained from its stated equation; the charges, momenta and L definition of its event are not given.
- **Figure 7.** Our hits are faster (median βγ about 2×); the paper's dE/dx panel is not reproduced from its own βγ panel by four standard definitions; thickness and selection are not stated.
- **Repeated layer crossings.** About half of our fiducial events cross some layer more than once; the paper assumes once. We use the first crossing.
- **Figure 6 coverage.** Λ > 25 keV not computed (integrator step ceiling); low-Λ edges at light masses not resolved (larger samples needed).
- **Colored f_{ch}.** Not stated whether the paper's Fig. 6 includes the both-charged factor.

Conclusion

- **Paper's method (Part A).** The trigger efficiency (colored), the fiducial efficiencies, the flat quirk planes and the background-free signal region are reproduced. The reconstruction efficiency is 3–8 times lower than Table I, and the Figure 6 reach is lower: colored 1497 / 1424 GeV (paper 1793 / 1707), EW 524 / 473 GeV (paper 762 / 688).
- **Our additional study (Part B).** Without SCT z smearing the signal numbers come close to the paper (Table I mostly consistent; reach colored 1702 / 1591 GeV, EW 792 / 701 GeV), but the converted-photon background becomes 10–48 events at 300 fb^{-1}. No single setting reproduces both the paper's signal efficiency and its zero background.
- **EW trigger.** The paper's 0.10 matches an LO cross-section ratio, not the matched + shower count of the method it describes.
- **Figure 2.** With the paper's own equation, |L| returns to zero at every second meeting (also shown analytically in arXiv:2002.07503); the paper's curve does not. L ≈ 0 does not by itself mean annihilation; a worst-case estimate lowers the reach by only 17–34 GeV (Section B6).
- **Expected events (Table II).** At the two benchmark masses the expected signal is a few events or less in both settings.

Final remarks on the paper

- The idea of the paper is clear and works in our simulation: quirk pairs leave hits on a thin plane (Δs of tens of μm before SCT z smearing), random hits rarely do, and the contours of Figure 6 have the paper's shape.
- The method is described in enough detail to implement the plane fit (Eqs. 9–10), the seeding and growing of planes, and the signal-region cuts; we implemented them as written.
- The numbers cannot be reproduced from the stated settings alone. Several inputs that decide them are missing: the SCT z resolution actually used, the signal cross sections and K-factors, the meaning of $N_Q = 2$, how the trigger efficiency was obtained, the normalisation of the backgrounds, the event behind Figure 2, and the dE/dx definition of Figure 7.
- The paper's main conclusion, a nearly background-free search with ε_{reco} up to ~70%, holds in our simulation only for a z resolution much better than the stated 1.16 mm, and in that case the converted-photon background is not negligible without further cuts (the paper suggests a Δφ(MET, n_3) or dE/dx requirement, which it does not apply).

References

Paper: S. Knapen, H. K. Lou, M. Papucci, J. Setford, Phys. Rev. D 96, 115015 (2017), arXiv:1708.02243. [27] ATLAS, ATL-PHYS-PUB-2009-002.
J. Li, T. Li, J. Pei, W. Zhang, “The quirk trajectory”, arXiv:2002.07503 (Eqs. II.11–II.14, A.4–A.5; Sec. VI, App. D).
J. Li, T. Li, J. Pei, W. Zhang, arXiv:1911.02223 (EW trigger 0.1 as an LO ratio, App. A). J. Li et al., arXiv:2108.06748. J. L. Feng et al., arXiv:2404.13814.
Particle Data Group, C. Patrignani et al. (2016), Passage of particles through matter, Eq. 34.12 (most probable energy loss) and Table 34.4 (silicon).
HSCP and monojet limits in Fig. 6: M. Farina, M. Low, private communication to the paper's authors (related: arXiv:1703.00912); taken here from the paper's figure.

Data and scripts: R_data_2026_10_15.py (all numbers), R_report_2026-10-21_partAB.py, F38–F42, F6 (F6_07 for the Part A / Part B split), F7 figures; Table II as tables/table_II_expected_events.csv; L tables (10^8 ħ) in L_VALUES_CSV_2026-10-21/; all plotted numbers as CSV in ROOT_CSV_FIGURES_2026-10-21/; ROOT versions of Figs. 2–7 in ROOT_GENERATED_GRAPHS/.